

Beyond Carrying Capacity: An Agent-Based Model for the Char Dham Pilgrimage Corridor

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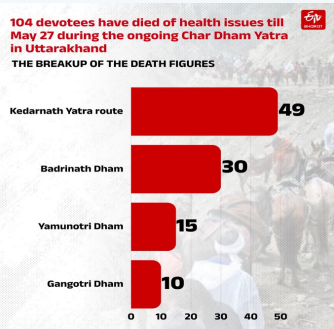
28-09-2026



Char Dham Yatra: Rising Demand and Infrastructure Response

EVIDENCE FROM RECENT MEDIA REPORTS

1 Demand surge + health risk



Deaths and crowding reports signal safety pressure before peak travel.

2 Corridor bottlenecks



Long jams in narrow valleys make reliability a transport-planning problem.

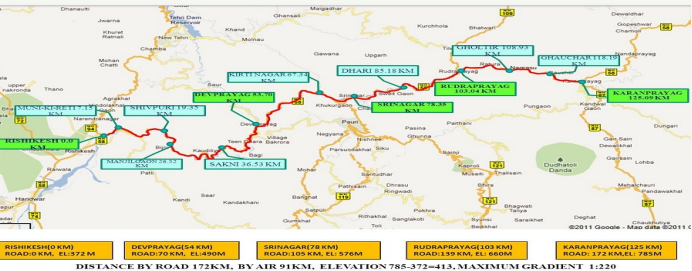
3 Record pilgrim registration

Record Crowds Overwhelm Char Dham Yatra Routes in Uttarakhand

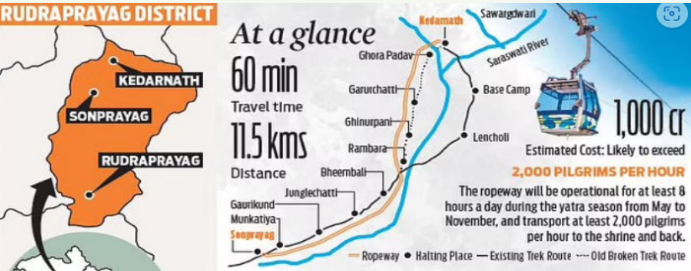
Last updated May 31

Over 1.5 million devotees visited Kedarnath, Badrinath, Gangotri, and Yamunotri in the first 30 days of the 2026 yatra, shattering records with 70,000-80,000 daily arrivals. Improved roads and government promotions fueled the surge, but it caused gridlock, stalled treks, and long queues for ponies and darshan. Authorities introduced token systems and one-way traffic, while environmentalists warn of risks to the fragile ecosystem, including Joshimath's subsidence and at least 40-50 pilgrim deaths from exhaustion or accidents.

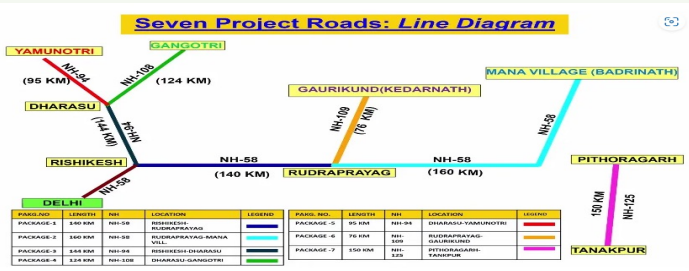
PROPOSED INFRASTRUCTURE DEVELOPMENTS



Rishikesh-Karanprayag Rail Marg



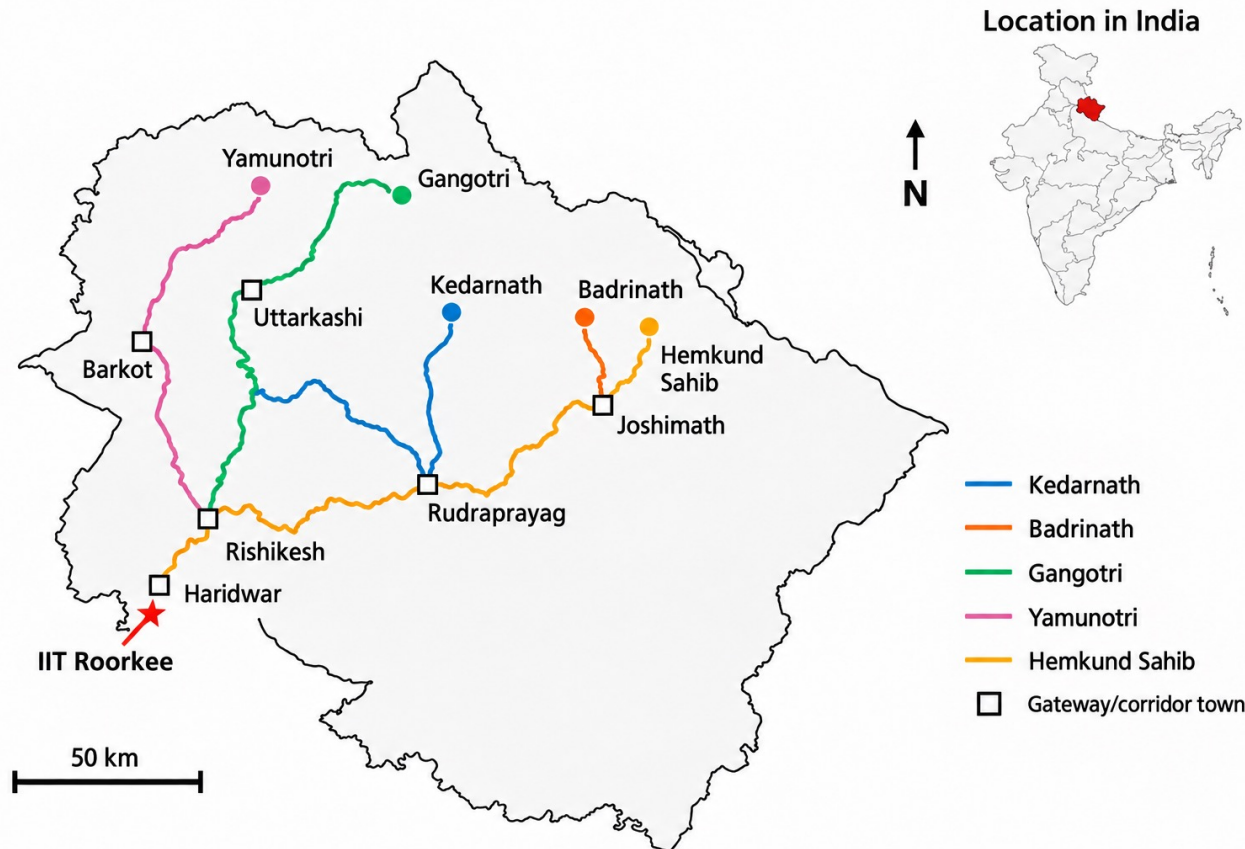
Gaurikund-Kedarnath Ropeway



Char Dham Mahamarg / Seven Project Roads

Source: [Web Sources](#) (accessed 04 June 2026)

Study Area: Char Dham, Uttarakhand

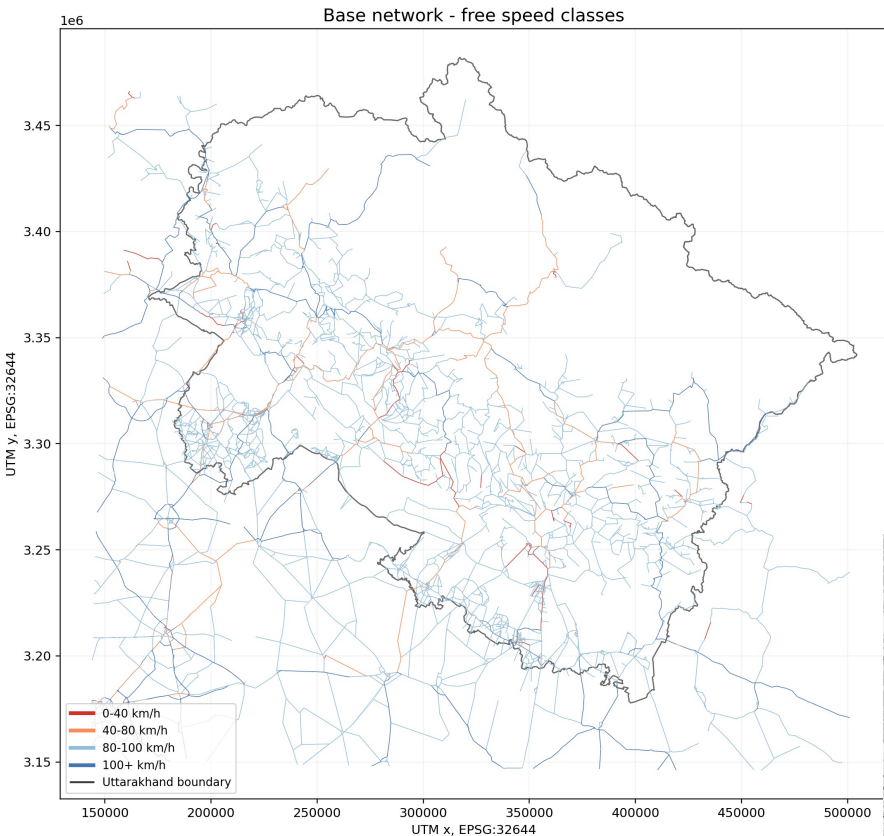
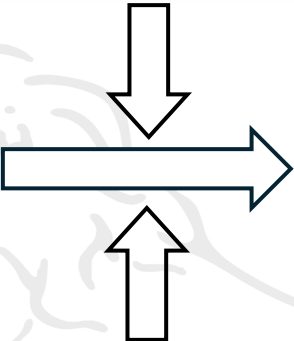


Schematic diagram: Char Dham facilities in Uttarakhand

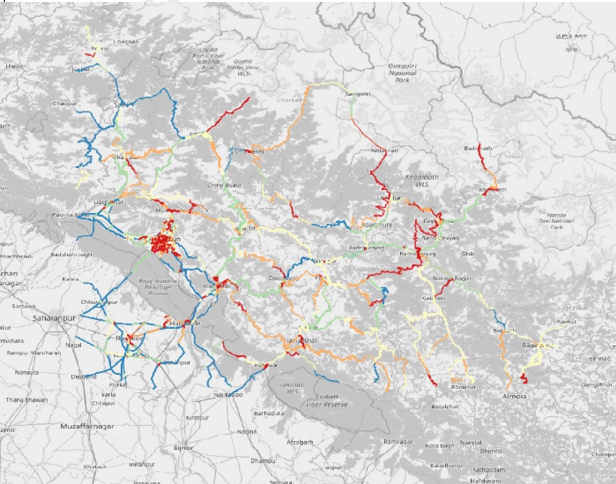
Typical pilgrim movement across the Char Dham route

Network Modification

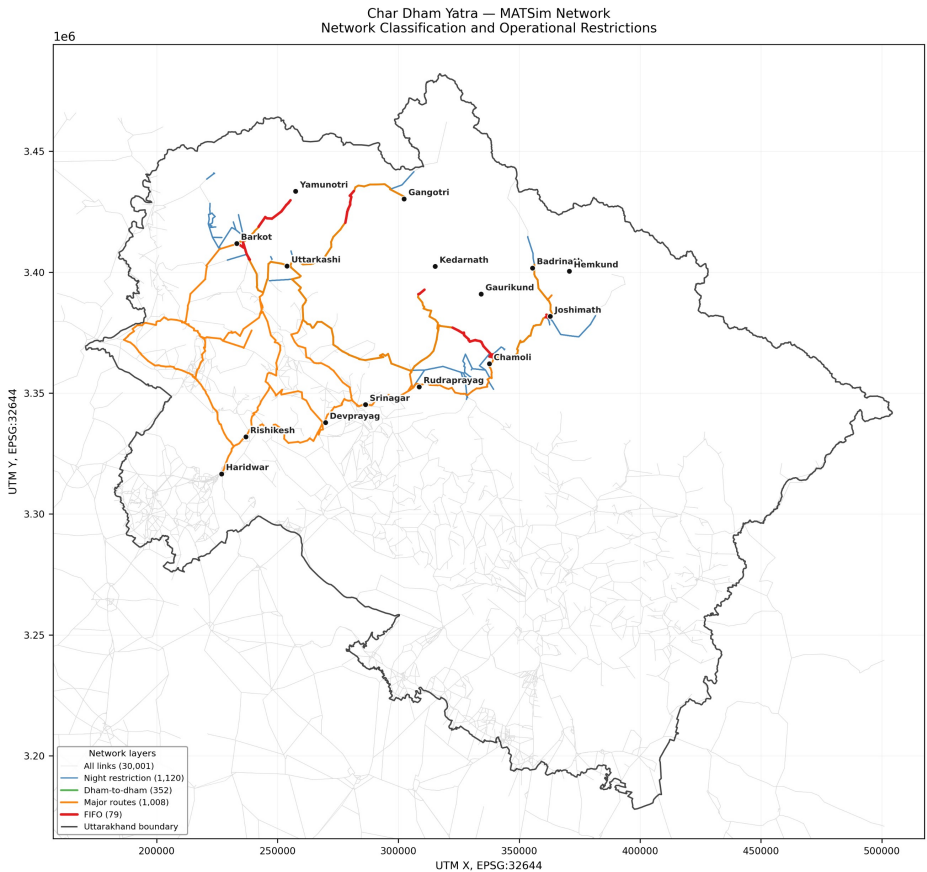
dham_2_dham contains the major routes taken from Dham to Dham
fifo_links the links having intermediate links
major routes contain the major routes taken from Haridwar to Dham



Network from OSM

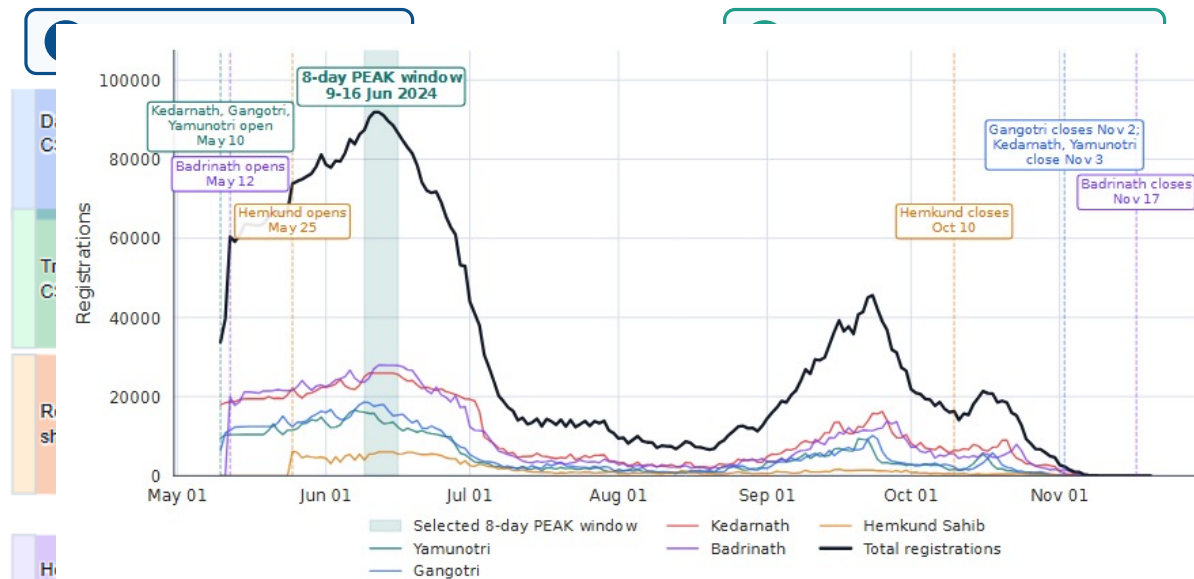


Here Map Speeds



Final Network

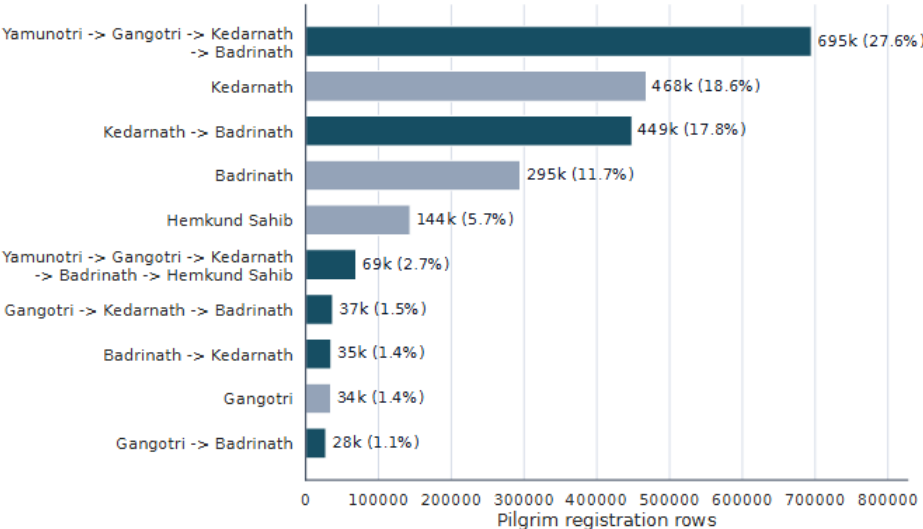
Registration Data to Plans



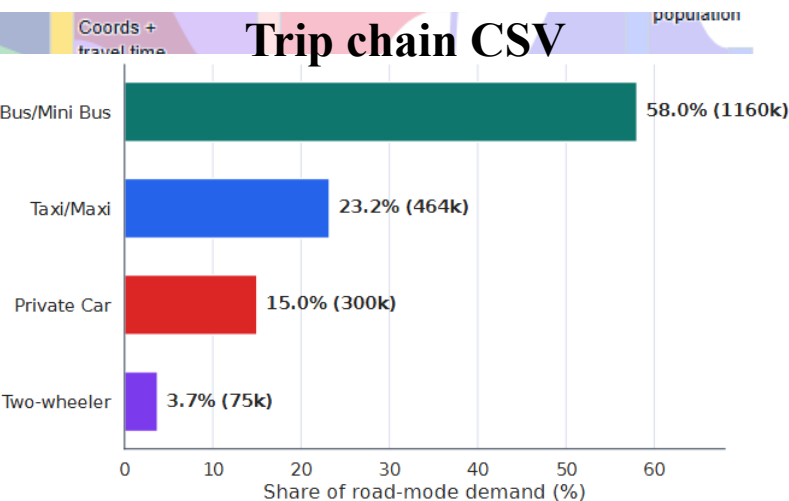
Registration Data

Simulation day	Date	Demand	Raw weight
1	09-06-2024	87,376	0.122
2	10-06-2024	90,422	0.126
3	11-06-2024	91,840	0.128
4	12-06-2024	91,858	0.128
5	13-06-2024	91,057	0.127
6	14-06-2024	89,598	0.126
7	15-06-2024	88,524	0.123
8	16-06-2024	86,660	0.121
Total	8 Days	717,335	1.000

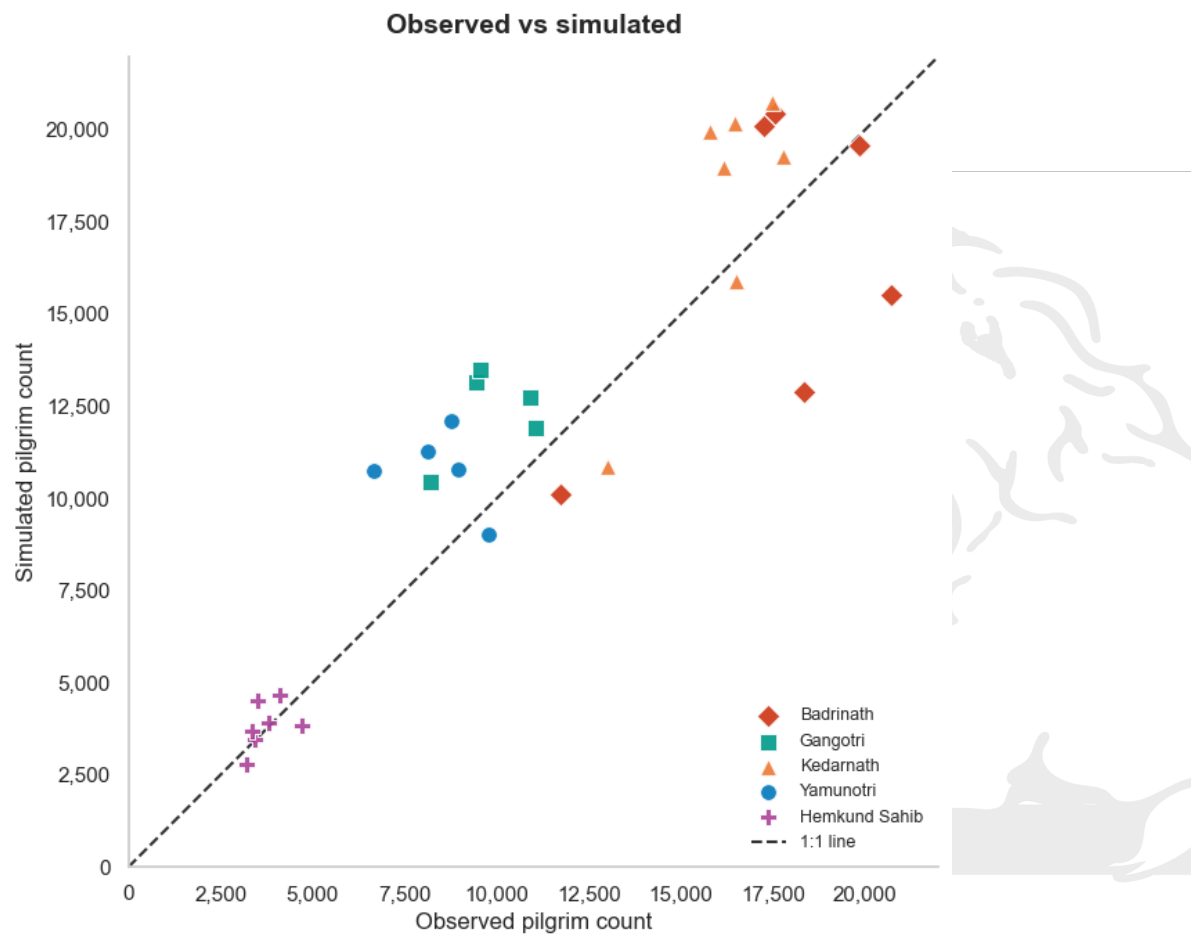
Rolling Window



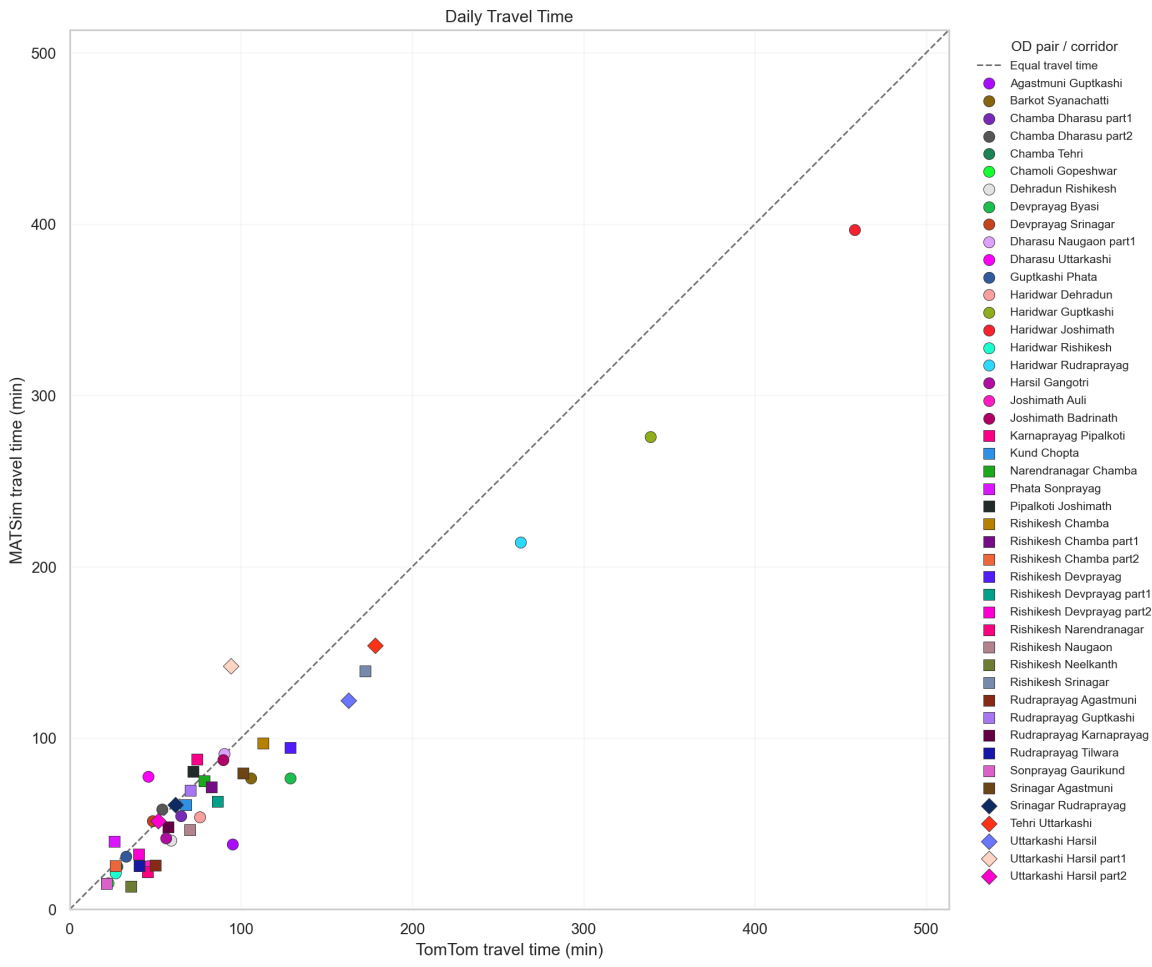
Trip chain CSV



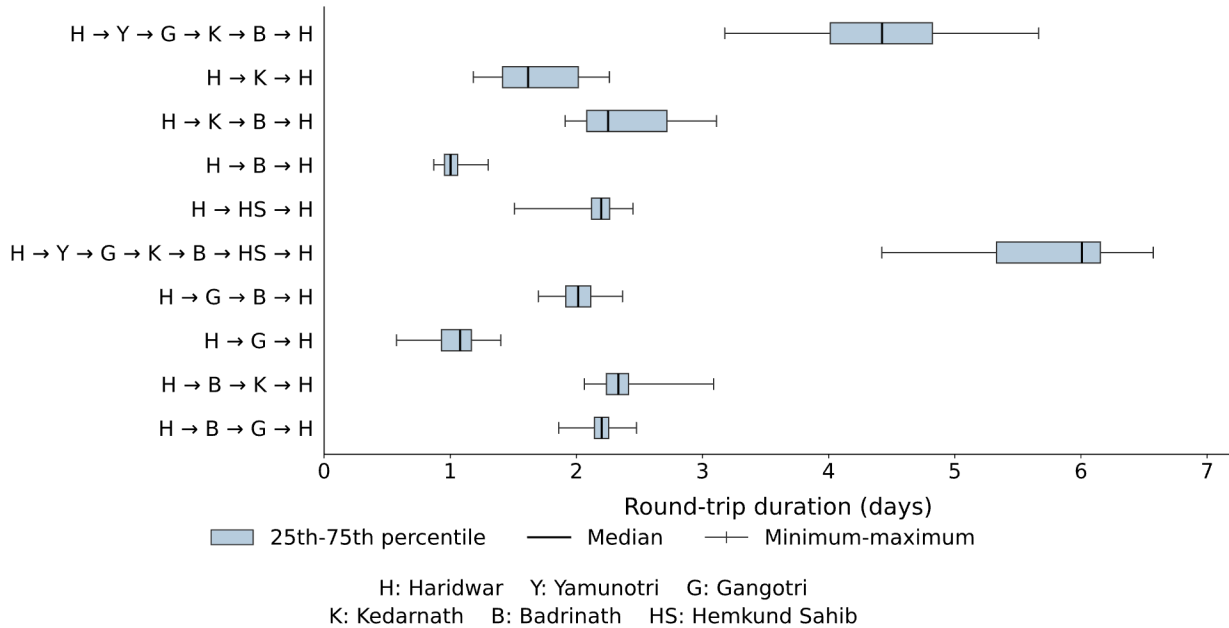
Road Mode Share



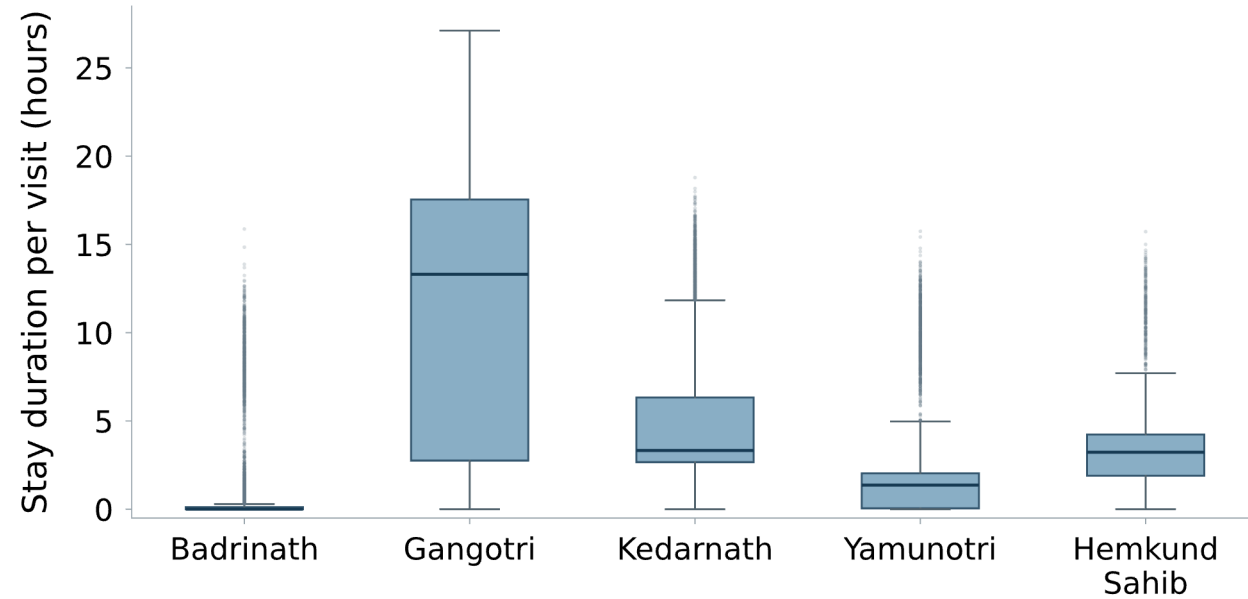
Simulated vs Observed Pilgrim Counts



Simulated vs Observed Travel Time
Bw The Corridors

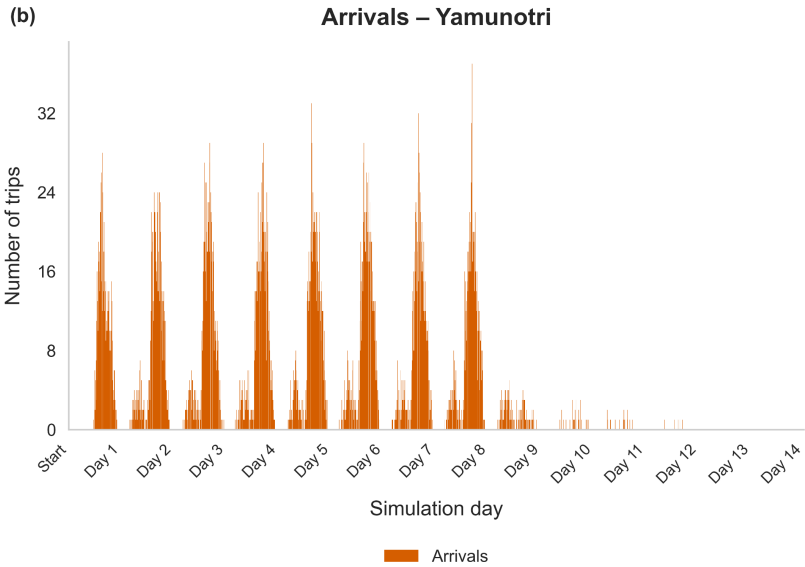
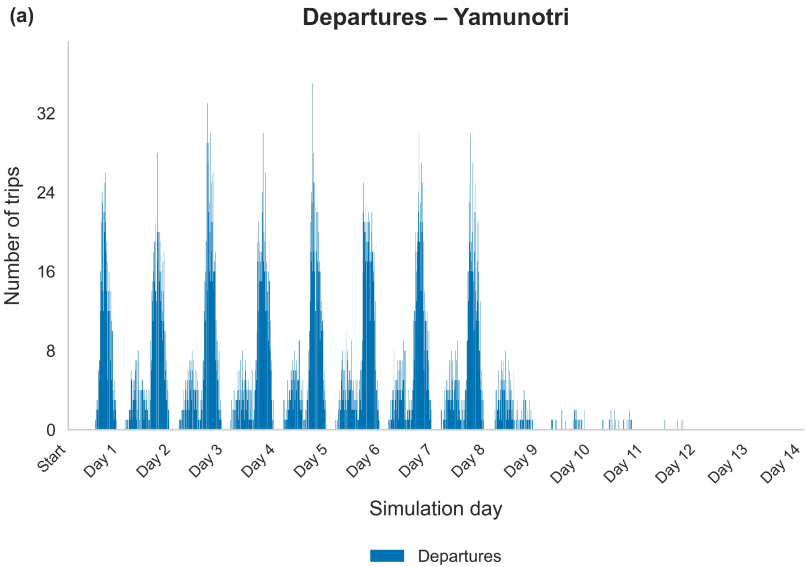
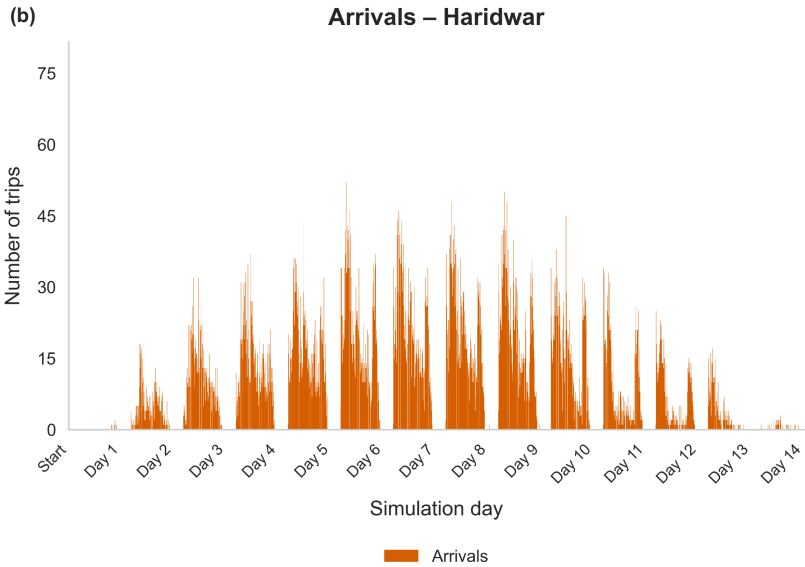
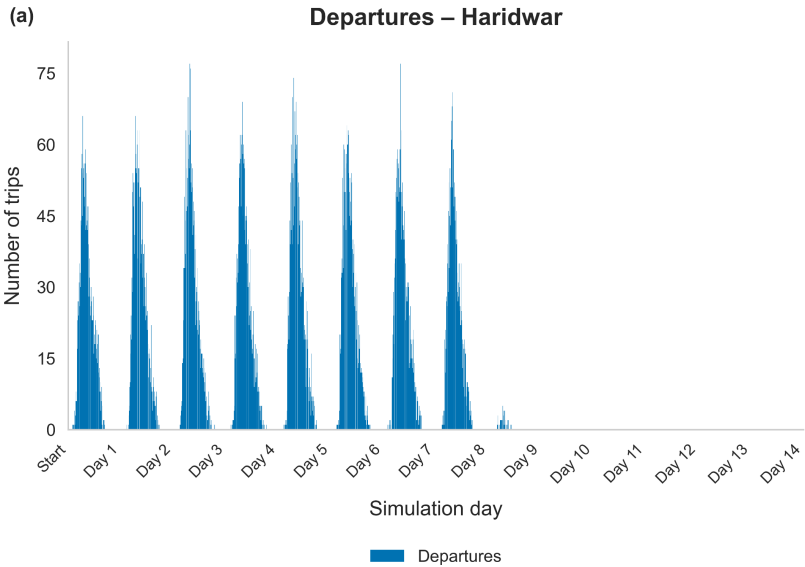


Pilgrim Trip Chain Duration

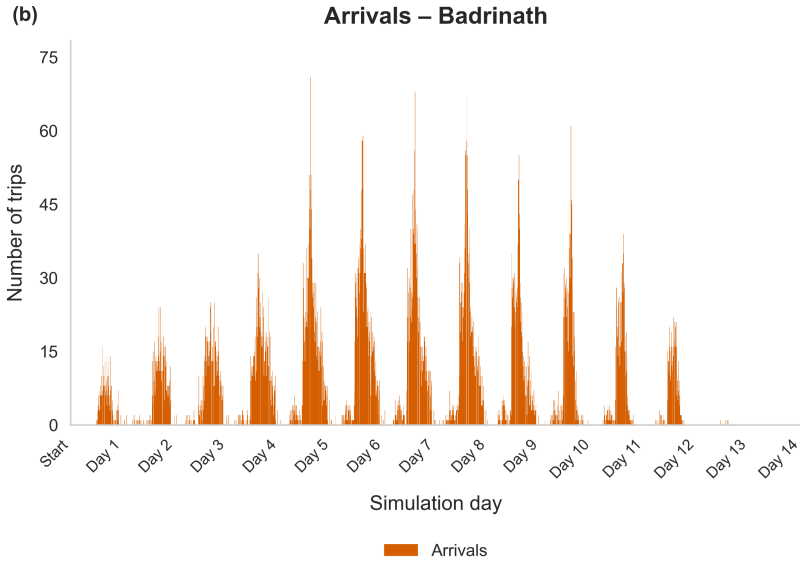
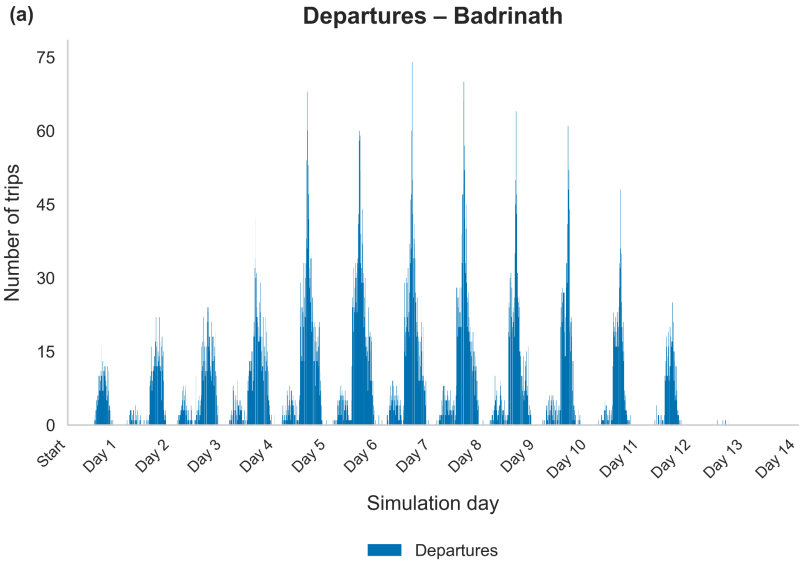
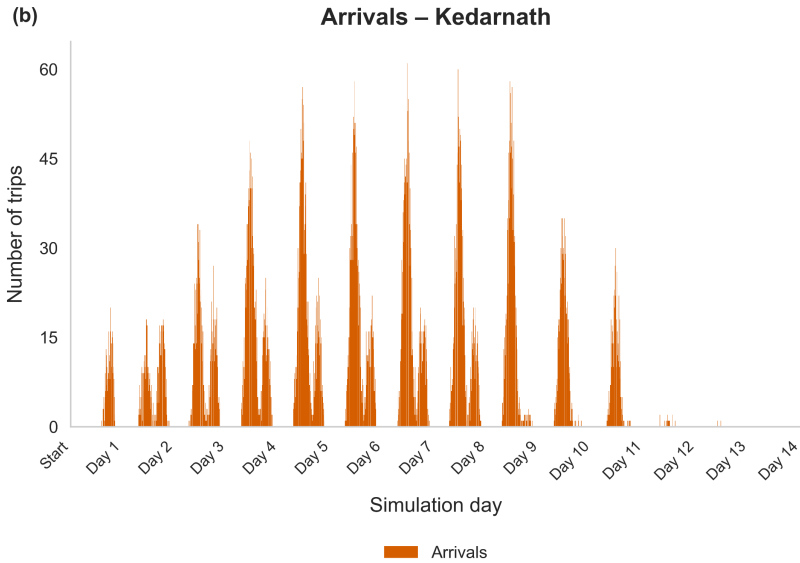
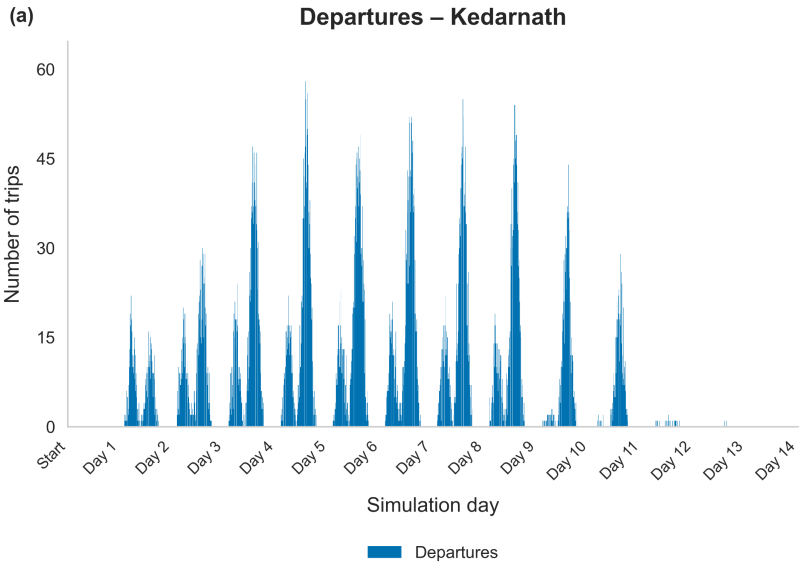


Shrine Stay Duration

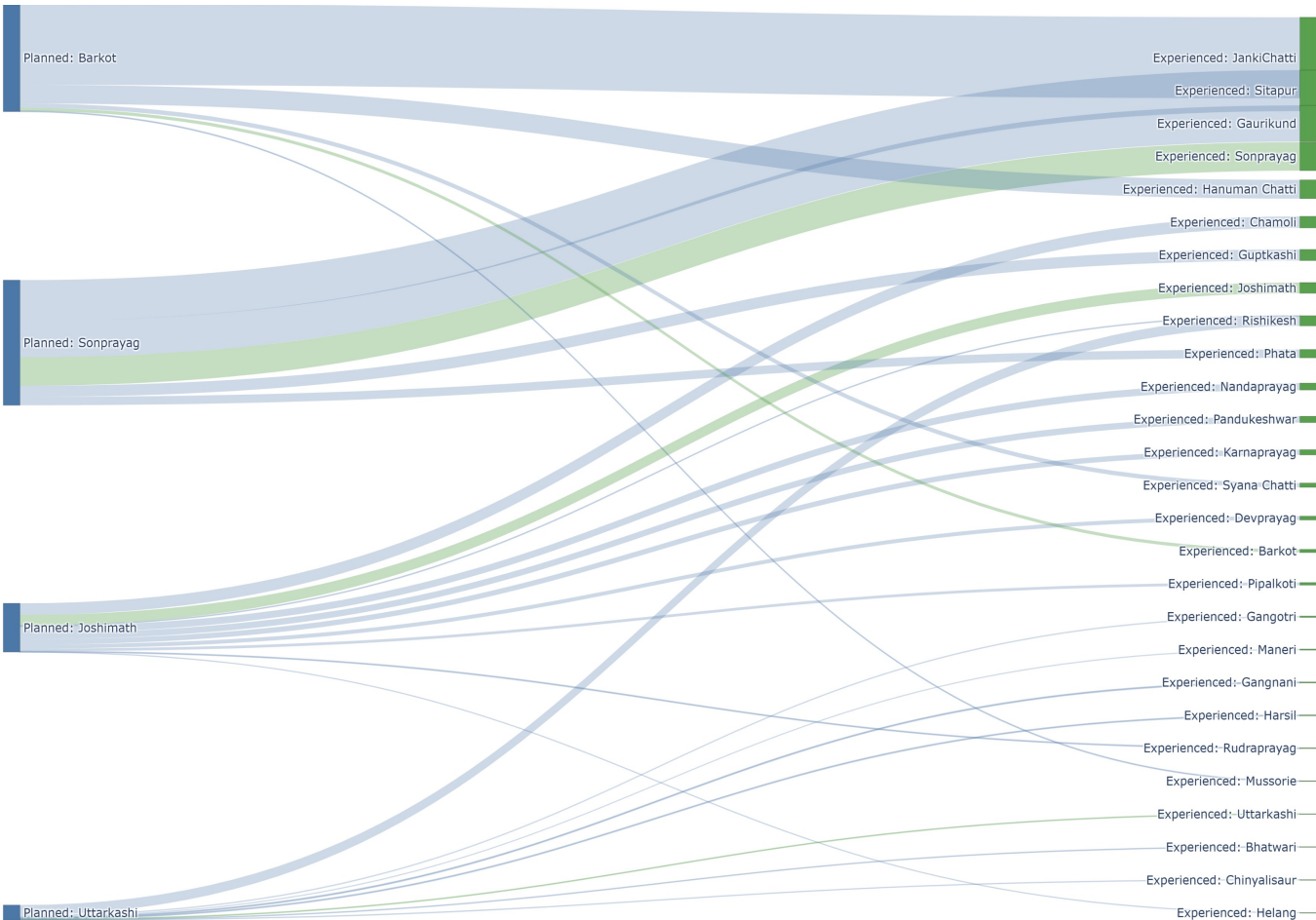
Arrival and Departure Plots



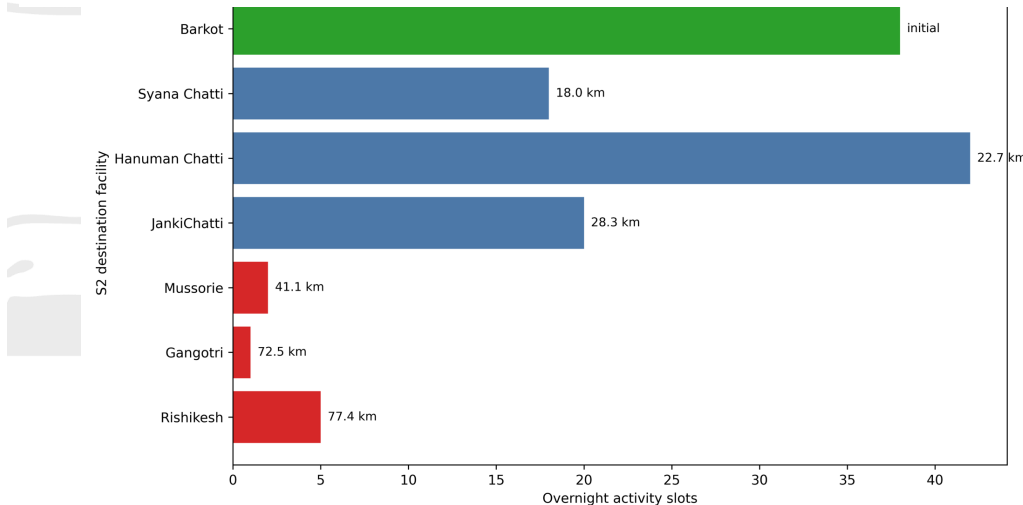
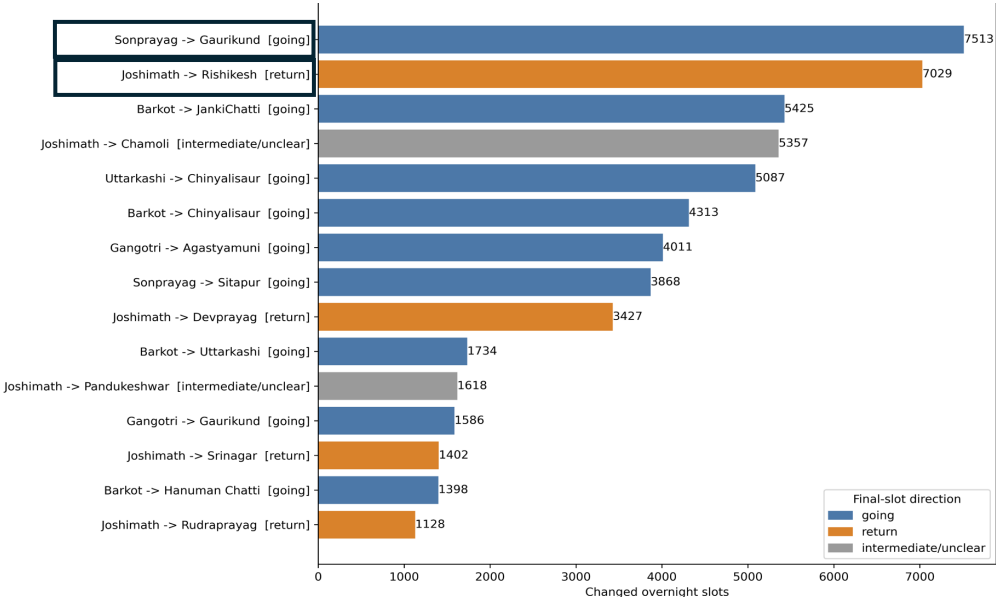
Arrival and Departure Plots



Reallocation To Nearby Facility : Location Choice



A typical example of how users reallocates



Relocation : Barkot Example

